

Supplementary Material

A Multitask Bayesian Optimization framework to identify a Gurson–Tvergaard–Needleman constitutive model for a MS1500 steel in hydrogen environment

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1 Useful schemes for BO approach

In Fig. 1, it is shown that after the GP training, it is possible to obtain an estimation of the tensile curve for an unknown pair of GTN parameters.

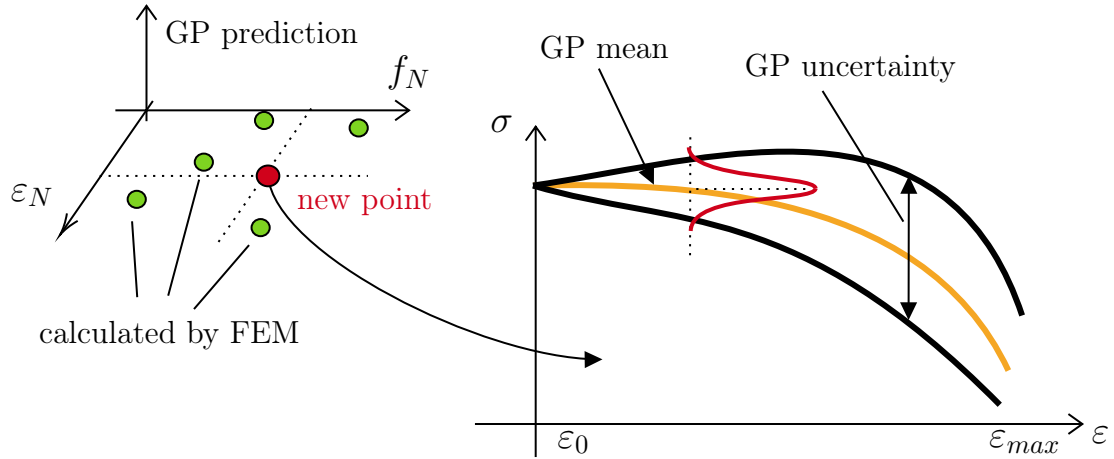


Figure 1: Every couple of GTN parameter, after GP training, is linked to a curve with its statistical uncertainty.

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2 Details on BO methods for GTN model parameters

Fig. 2 illustrates the values of the acquisition function for $C_d=0$ at the initial step with the five initial FEM simulations (on the left) and after the first iteration (on the right). In the left image, the search area for the minimum is highlighted as the lower-left region, colored in blue. It should be noted that the acquisition function was computed using the UCB method and incorporates information on both the mean and variance of the loss function. The minimum of this function within the defined domain corresponds, as discussed above, to the pair of GTN model parameters that best reproduce the experimental curve.

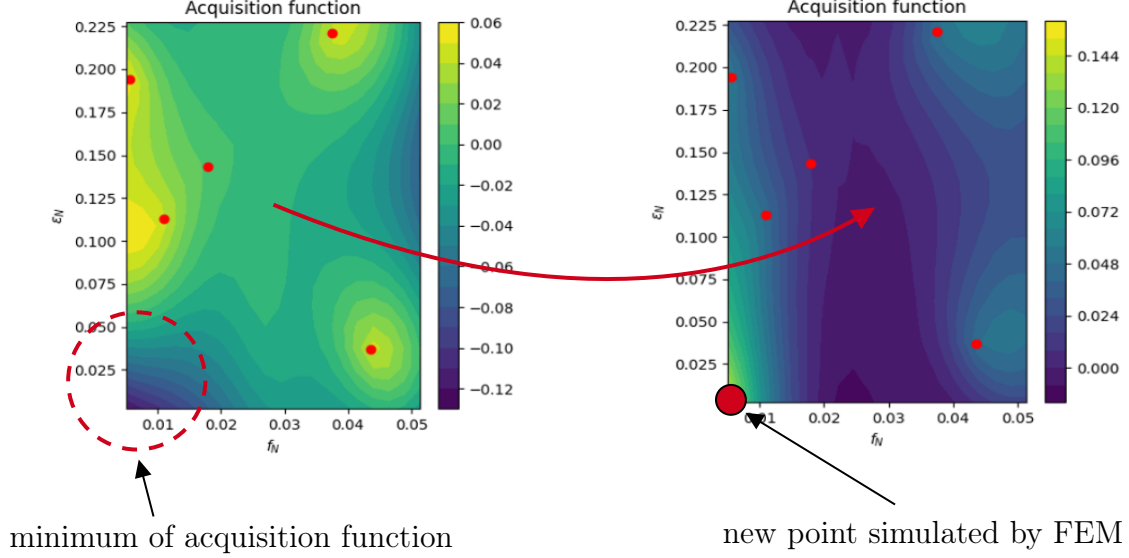


Figure 2: Choice of the new pair of GTN parameter f_N and ϵ_N for an additional FEM simulation.

Each time a new minimum point is identified, an additional FEM simulation is performed, providing updated and more accurate estimates from the GP. As the GP is updated, the loss function changes, which in turn modifies the acquisition function. The right panel shows the acquisition function after the new point has been simulated, illustrating a significant shift in the minimum position. This indicates that the new region to search for the optimal parameters lies around the intermediate values within each interval.

3 Additional plots and data

Here are all the graphs obtained for the six simulations conducted starting from a reduced initial database for GP training. In each table, for each graph, the simulated curve and the number of iterations required to obtain a loss of less than 10 N are reported.

3.1 Group S1

See Table 1.

3.2 Group S2

See Table 2.

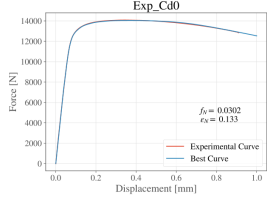
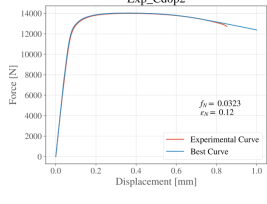
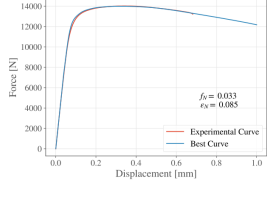
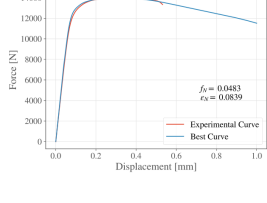
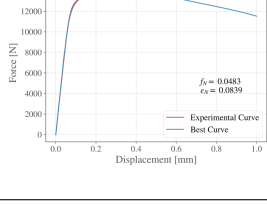
C_d	Loss and Iterations	Plots (Group S1)
0.0	Loss = 9.07 iter = 7	
0.2	Loss = 8.43 iter = 3	
0.5	Loss = 4.34 iter = 1	
0.7	Loss = 3.64 iter = 1	
0.8	Loss = 2.77 iter = 1	

Table 1: Values of Loss and number of iterations for each simulation in group S1, and comparison between experimental and simulated curves.

3.3 Group S3

See Table 3.

3.4 Group S4

See Table 4.

3.5 Group S5

See Table 5.

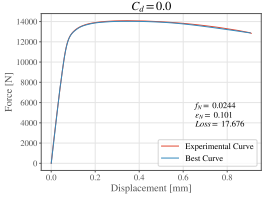
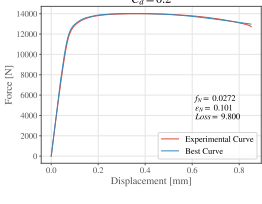
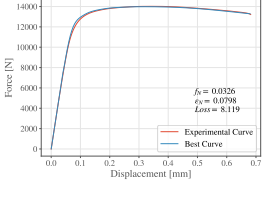
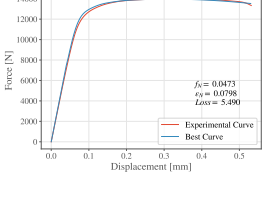
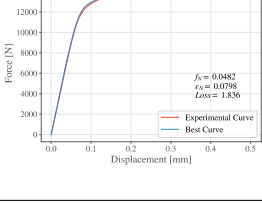
C_d	Loss and Iterations	Plots (Group S2)
0.0	Loss = 17.68 iter = 10	 $C_d = 0.0$ $f_c = 0.0244$ $e_2 = 0.101$ $Loss = 17.676$
0.2	Loss = 9.80 iter = 2	 $C_d = 0.2$ $f_c = 0.0272$ $e_2 = 0.101$ $Loss = 9.800$
0.5	Loss = 8.12 iter = 2	 $C_d = 0.5$ $f_c = 0.0326$ $e_2 = 0.0798$ $Loss = 8.119$
0.7	Loss = 5.49 iter = 1	 $C_d = 0.7$ $f_c = 0.0473$ $e_2 = 0.0798$ $Loss = 5.490$
0.8	Loss = 1.84 iter = 3	 $C_d = 0.8$ $f_c = 0.0482$ $e_2 = 0.0798$ $Loss = 1.836$

Table 2: Values of Loss and number of iterations for each simulation in group S2, and comparison between experimental and simulated curves.

3.6 Group S6

See Table 6.

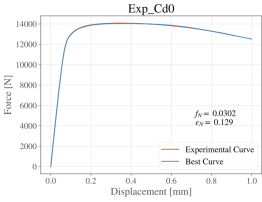
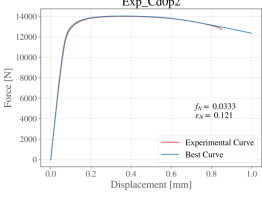
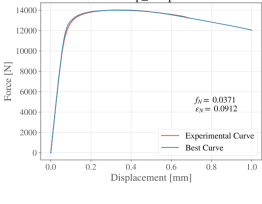
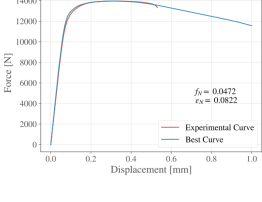
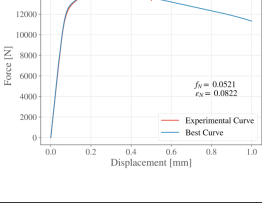
C_d	Loss and Iterations	Plots (Group S3)
0.0	Loss = 6.74 iter = 4	
0.2	Loss = 8.27 iter = 1	
0.5	Loss = 4.80 iter = 3	
0.7	Loss = 3.94 iter = 2	
0.8	Loss = 2.27 iter = 1	

Table 3: Values of Loss and number of iterations for each simulation in group S3, and comparison between experimental and simulated curves.

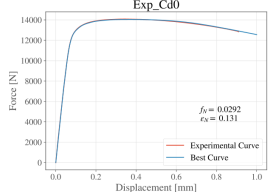
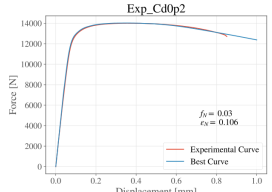
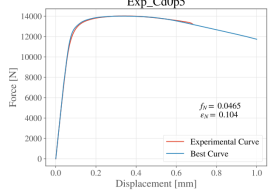
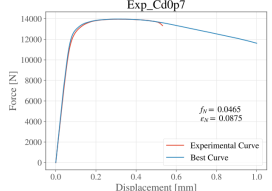
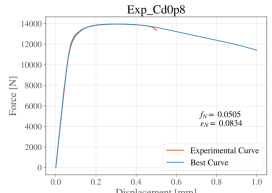
C_d	Loss and Iterations	Plots (Group S4)
0.0	Loss = 9.97 iter = 10	
0.2	Loss = 9.83 iter = 2	
0.5	Loss = 5.30 iter = 1	
0.7	Loss = 2.12 iter = 1	
0.8	Loss = 1.48 iter = 1	

Table 4: Values of Loss and number of iterations for each simulation in group S4, and comparison between experimental and simulated curves.

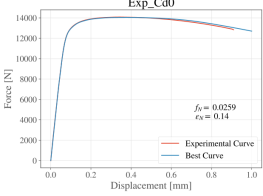
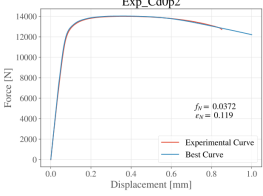
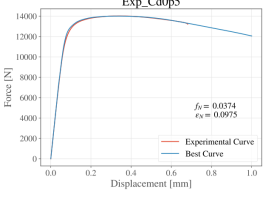
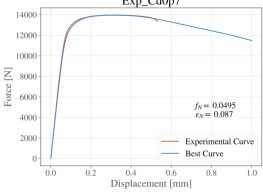
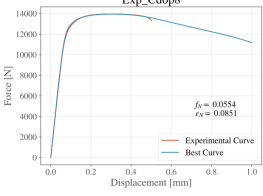
C_d	Loss and Iterations	Plots (Group S5)
0.0	Loss = 24.0 iter = 10	
0.2	Loss = 9.79 iter = 2	
0.5	Loss = 3.32 iter = 1	
0.7	Loss = 2.75 iter = 1	
0.8	Loss = 2.12 iter = 1	

Table 5: Values of Loss and number of iterations for each simulation in group S5, and comparison between experimental and simulated curves.

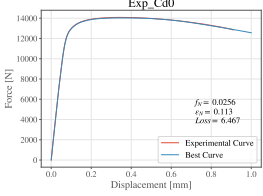
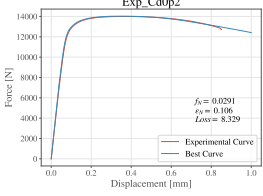
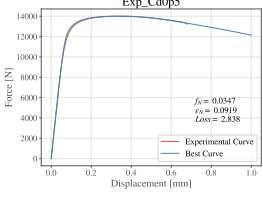
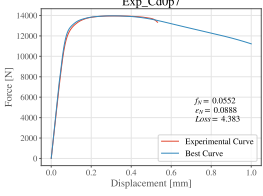
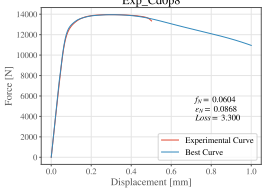
C_d	Loss and Iterations	Plots (Group S6)
0.0	Loss = 6.47 iter = 1	 Exp_Cd0 Force [N] Displacement [mm] Experimental Curve Best Curve $f_c = 0.0256$ $e_2 = 0.113$ Loss = 6.407
0.2	Loss = 8.33 iter = 6	 Exp_Cd0p2 Force [N] Displacement [mm] Experimental Curve Best Curve $f_c = 0.0291$ $e_2 = 0.106$ Loss = 8.329
0.5	Loss = 2.84 iter = 1	 Exp_Cd0p5 Force [N] Displacement [mm] Experimental Curve Best Curve $f_c = 0.0347$ $e_2 = 0.0919$ Loss = 2.838
0.7	Loss = 4.38 iter = 1	 Exp_Cd0p7 Force [N] Displacement [mm] Experimental Curve Best Curve $f_c = 0.0552$ $e_2 = 0.0888$ Loss = 4.383
0.8	Loss = 3.30 iter = 1	 Exp_Cd0p8 Force [N] Displacement [mm] Experimental Curve Best Curve $f_c = 0.0604$ $e_2 = 0.0868$ Loss = 3.300

Table 6: Values of Loss and number of iterations for each simulation in group S6, and comparison between experimental and simulated curves.